# Matthew Emmanuel T. Labrador

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## EDUCATION

**University of the Philippines Manila** *Manila, PH*

*BS Computer Science, University Scholar | Expected May 2029*

Current Cumulative GWA: 1.0375 (Summa Cum Laude standing) • DOST Undergraduate Scholar

**PAREF Southridge School** *Manila, PH*

*Senior High School, Class Valedictorian and Gold Medalist | May 2025*

Mercury Drug 2025 Gawad Talindaw Award for Excellence in Science

## EXPERIENCE AND PROJECTS

**NVIDIA Philippines | Campus Ambassador** *2026–Present*

**PC Express | Intern** *2026–Present*

- Across these roles, develop local AI applications with NVIDIA RTX 5090 GPUs and deliver talks and training on using and deploying models on personal hardware.

**Tumbang Preso (TUMP) | Gameplay & Systems Engineer** *2026*

*Godot 4, GDScript, Blender, ENet UDP Authoritative Netcode, Dedicated Linux Servers*

- Won 1st Place Champion with Team BH Studios at Gear Up NCR Esports Game Dev Challenge; selected as NCR Representative to National Finals (backed by DOST & Level Up).
- Engineered core multiplayer systems from empty project to shipped build in 5 days: gameplay mechanics, 3D asset pipeline, sound design, bot AI, and netcode.
- Engineered authoritative-host netcode over ENet with distance-based collision validation after profiling and discovering 16 of 36 built-in engine physics overlap callbacks silently dropped under network jitter.
- Deployed dedicated online lobbies on a Linux VPS with UDP discovery and join codes, reaching players across different networks and CGNAT.

**eGovMed & PGH OCRA Mobile | Full-Stack Lead & Systems Architect** *2026*

*React 18/19, Vite, Capacitor (Android/iOS), Node.js, Express, TypeScript, Redis, Server-Sent Events, 8 eGovPH APIs*

- Grand Champion at eGov Hackathon PH 2026 (₱100,000 prize), with Philippine General Hospital as the pilot target.
- Engineered cross-platform mobile app & PWA featuring bilingual AI triage with deterministic safety floors (100% emergency recall, 0% false alarms).
- Architected real-time queue stream over SSE with dynamic rolling-median ETAs and cross-instance Redis fan-out (load-tested to 6,000 concurrent streams at <600ms latency); integrated 8 national eGovPH APIs with atomic Redis compare-and-set liveness locks.
- Authored 127 automated security & safety tests, envelope encryption under Data Privacy Act, and VAPID web push notifications.

**Knee MRI Reader | Deep Learning & Computer Vision Engineer** *2026*

*Python, PyTorch, DINOv2 (ViT), Cross-View Attention, ONNX Runtime, Next.js 16, React 19, TypeScript*

- Trained an end-to-end diagnostic model scoring 12 knee pathologies across 4,407 DICOM MRI studies, achieving 0.843 Out-of-Fold Macro ROC-AUC (0.96 Medial Osteoarthritis, 0.95 Effusion/Baker's Cyst, 0.90 Fracture, 0.88 ACL).
- Designed multi-view cross-attention architecture: mapped varying patient series into 6 canonical anatomical planes using directional cosine vectors, extracting frozen DINOv2 patch features with 12 specialized diagnostic attention heads.
- Built a 3-LLM weak supervision consensus engine on 4,349 multi-lingual free-text reports, boosting label accuracy from 0.87 to 0.89.
- Deployed real-time workstation: exported a 20-model ensemble to sub-200ms ONNX models with interactive client-side DICOM slice parsing.

**Diabetic Complication Swarm (GlycoSwarm AI) | Multi-Agent Systems Lead** *2026*

*Python, LangGraph, FastAPI, Ollama, Gemma 4 26B, Fireworks GLM 5.2, AMD MI300X GPU, Next.js*

- Lead developer of international team at AMD Hackathon ACT II building a LangGraph StateGraph of 4 parallel specialist agents (Renal, Neuropathy, Retinal, Cardiovascular) that dynamically generate and execute Python scoring scripts against real NHANES lab data.
- Served live LLM inference on AMD MI300X accelerators via Ollama with automated failover routing to Fireworks GLM.

**Khanpanion | Team Lead & Developer** *2026*

*Next.js, React, TypeScript, Khan Academy Resources, Rule-Based Remediation*

- Grand Champion, Khan Academy Education Impact Challenge 2026 (Enactus Philippines).
- Developed a Next.js study platform with typed prerequisite routing, generated practice, resumable mock exams and device-local learning state, linked to targeted Khan Academy resources.

**Heart Disease Prediction Model | Machine Learning Engineer** *2025*

*Python, scikit-learn, pandas, Streamlit, Google Colab*

- Trained a supervised machine learning classification pipeline in Python to predict cardiovascular disease risk from patient clinical features end-to-end.
- Performed data preprocessing, missing value imputation, and categorical feature engineering using pandas and scikit-learn.

## TECHNICAL SKILLS

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|---|---|
| **Languages:** | Python, SQL, C, C++17, TypeScript, JavaScript, GDScript, HTML |
| **ML & Data:** | Feature Engineering, Evaluation Metrics (ROC-AUC, Precision, Recall, F1), pandas, Prompt Engineering, Model Deployment |
| **Agentic AI & Orch:** | PyTorch, DINOv2 (ViT), ONNX Runtime, LangChain, LangGraph, Multi-Agent Systems, FastAPI, Ollama, RAG, SHAP |
| **Web & Mobile:** | Next.js 16, React 19/18, Node.js, Express, Vite, Tailwind CSS, Capacitor (Android/iOS), PWA, Redis, SSE, VAPID Web Push |
| **Systems & Tools:** | Godot 4, ENet Authoritative Netcode, Raylib, CMake, WebAssembly, Blender, Linux VPS, Git/CI-CD (GitHub Actions) |
| **Certifications:** | AMD Multiagent Systems Deployment, CS50 Intro to CS, AI Prompt Engineering, Python, C |

## LEADERSHIP AND ACTIVITIES

**Competitive Debater** *2022–2025*

Quarterfinalist at Ateneo Peace Debate; competed at NASH DC (DLSU), PSDC (Ateneo), UP Diliman Debates, XSDC, and ASDC.

**Math Team Captain — PAREF Southridge School** *2022–2024*

Led tutoring sessions and mentored student peers for local and regional inter-school mathematics competitions.

**Google Developer Group & UP Socomsci** *2025–Present*

Coordinated cross-functional student developer teams on operations and event setup for tech conferences and workshops.

## AWARDS

Grand Champion, Khan Academy Education Impact Challenge (2026) • Grand Champion, eGov Hackathon PH (2026, ₱100,000) • 1st Place, Gear Up NCR (2026, NCR Representative to Nationals) • DOST Undergraduate Scholar (2025–Present) • Top 5 Finalist, Olymphysics NCR (2025) • 5th Place, Philippine Statistics Quiz NCR (2025)